

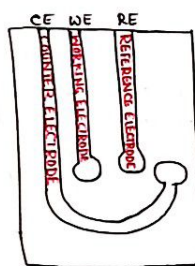
Explain the electrochemical sensors used for the detection of Hydrocarbons

- * Among the various dangerous hydrocarbon pollutants, polycyclic aromatic hydrocarbons (PAHs) are widely found in air, water, soil and food.
- * Hence detection of PAHs is essential to monitor their toxicity and carcinogenic risk.
- * After entering the body, some PAHs are metabolized into 1-hydroxypyrene which is commonly found in urine sample.
- * Electrochemical sensors are used for the detection of 1-hydroxy pyrene in urine sample.

Construction

- * It contains a plastic strip on which three electrodes are printed
- * Working electrode is made up of polyamidoamine and chromium centered metal organic framework, Electrically reduced graphene oxide and glassy carbon Electrode.
- (or)
Working electrode: PAMAM / Cr-MOF / ERGO / GCE
- * Counter electrode: platinum
- * Reference electrode: - Ag / AgCl
- * Upper portion is sealed with insulating material and strip is connected to electrical connector and immersed in a analyte solution for measurement

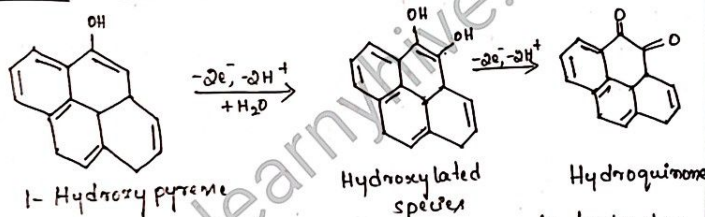
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Working

- * The working is designed such that 1-Hydroxy pyrene selectively get absorbed on its surface.
- * On immersing the sensor in analyte solution followed by applying the suitable voltage via reference electrode, the oxidation

of 1-Hydroxy pyrene takes place on the surface of sensing electrode
Oxidation reaction is



- * The electrons liberated at the working electrode due to the oxidation of 1-Hydroxy pyrene will result in the flow of current between counter electrode and working electrode
- * The number of liberated electrons is proportional to the concentration of 1-Hydroxy pyrene in the analyte solution
- * By measuring the current, the concentration of 1-Hydroxy pyrene can be determined.

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Q. Explain the detection of Ascorbic acid by using a Electrochemical sensors.

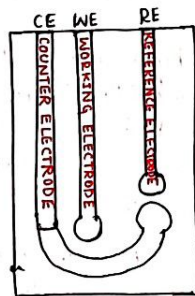
- * Ascorbic acid is also known as vitamin C, is a water soluble vitamin derived from green vegetables, fruits and other dietary supplements. It is well known anti-oxidant and essential nutrient in human diets and is also used as a food preservative.
- * Ascorbic acid improves the immune system, enables collagen synthesis, which is needed to maintain healthy bones, teeth, skins, Cartilage.
- * Abnormal ascorbic acid levels in body fluids may cause cancer, cardiovascular diseases and Alzheimer's and Parkinson's disease.
- * The detection of ascorbic acid is important in the field of food safety and medical diagnosis.

Construction

It contains a plastic polyester sheet as a strip on which three electrodes are printed. Working electrode and counter electrode is made up of multi walled carbon nanotubes (MWCNT) modified with gold nanoparticles and Reference electrode is made up of silver-silver chloride. Upper portion is sealed with insulating material, strip is connected to electrical connector and immersed in a analyte solution for the measurement.

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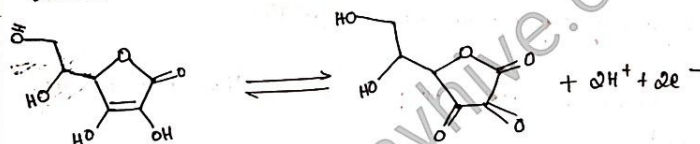
WORKING:-



* The working electrode is designed such that the ascorbic acid selectively get absorbed on its surface.

* On immersing the sensor in analyte solution followed by applying the suitable voltage via reference electrode, the oxidation of ascorbic acid takes place on the

Surface of sensing electrode
Oxidation Reaction is



Ascorbic acid

Dehydro Ascorbic acid

- * The electrons liberated at the sensing electrode will result in the flow of current between counter electrode and working electrode.
- * The number of liberated electrons is proportional to the concentration of ascorbic acid in the analyte solution.
- * By measuring the current, concentration of ascorbic acid can be determined.

Q8 Explain the detection of diclofenac by using a electrochemical sensors.

* Diclofenac is most frequently prescribed non-steroidal anti-inflammatory drugs with antipyretic and analgesic effects.

* Diclofenac widely used for the treatment of acute muscle pain arthritis and osteoarthritis.

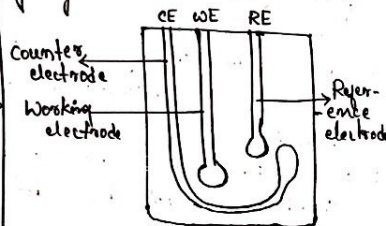
Construction

It contains a plastic strip on which three electrodes are printed using screen printing technology. The working electrode and counter electrode are printed with the carbon paste along with multiwall carbon nanotubes (or) Graphene and reference electrode is printed using Ag/AgCl electrode and the upper part of the sensor is sealed with the insulating material and the strip connected to electrical connector and immersed in the analyte solution for the measurement.

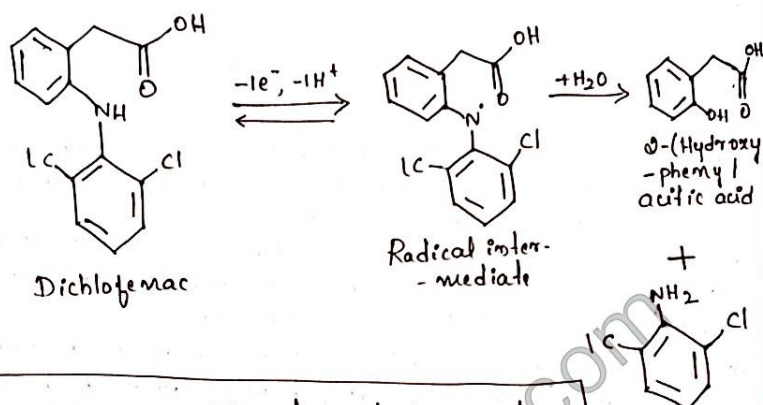
Working:-

* The working electrode is designed such that the Diclofenac selectively get absorbed on its surface.

* On immersing the sensor in analyte solution followed by applying the suitable voltage reference electrode oxidation of Diclofenac takes place.



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* The electrons liberated at the working electrode due to the oxidation of Diclofenac will result in the flow of current between counter electrode and working electrode.

* The number of liberated electrons is proportional to the concentration of Diclofenac in the analyte solution.

* Hence the concentration of Diclofenac is measured by measuring the current.

Materials for Memory and Display Systems

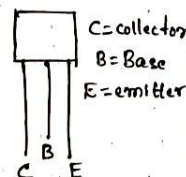
01. What are memory devices? Explain the classification of electronic memory devices with example.

Ans:- A memory device is a piece of hardware used to store the data and memory is stored in the form of zero's & one's.
ex:- CD, DVD, USB and external hard disc.

Classification of memory devices

(i) Transistor-type Electronic memory

* A transistor is a semiconductor device consists of at least three terminals for connecting the electric circuit.



* It can produce a stronger output signal, a voltage or current, proportional to weaker input signal, acting as an amplifier.

* It can be extensively used for switching operations, either for opening or closing a circuit.

* The transistor-type memory device consists of billions of transistors, each transistor represents one bit of information which is a 0 or a 1.

* When transistor is turned on, it represents a 1, and when it's turned off, it represents a 0.
ex:- DRAM.

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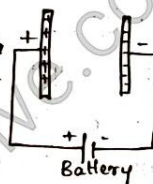
2. Capacitor-type electronic memory

* A capacitor is a two-terminal electrical device that can store energy in the form of an electric charge.

* It consists of two metal plates that are separated by an insulating material known as a dielectric.

* Charging and discharging are the two main effects of capacitors, which are used in capacitor-type electronic memory.

* The capacitor-type device consists of billions of capacitors, each capacitor represents one bit of information, which is a 0 or a 1.



* When a capacitor is charged, it represents a 1 and when it is discharged, it represents a 0.

ex:- Ferroelectric Random Access memory (FeRAM)

3. Resistor-type electronic memory

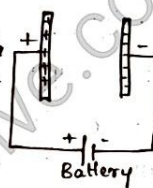
* The resistor is an electrical component that limits or regulates the flow of electrical current in an electronic circuit.

* The resistor switching devices are made up of metal-insulator-metal (MIM) structure in which the middle layer acts as an active switching medium.

change.

- * It consists of two metal plates that are separated by an insulating material known as a dielectric
- * charging and discharging are the two main effects of capacitors, which are used in capacitor-type electronic memory.

- * The capacitor-type device consists of billions of capacitors, each capacitor represents one bit of information, which is a 0 or a 1.



- * When a capacitor is charged, it represents a 1 and when it is discharged, it represents a 0.

ex:- Ferroelectric Random Access memory (FeRAM)

2. Resistor-type electronic memory

- * The resistor is an electrical component that limits or regulates the flow of electrical current in an electronic circuit

- * The resistor switching devices are made up of metal-insulator-metal (MIM) structure in which the middle layer acts as an active switching medium.

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02

- * The process of charging and discharging switches the device b/w low and high resistance states.

- * The resistor-type memory device consists of billions of resistors, each resistor represents one bit of information, which is a 0 or a 1.



- * Low resistance state represents a 1 and high resistance state represents a 0.

ex:- Programmable read only memory (PRAM)

3. Charge Transfer type electronic memory

- * A charge-transfer complex describes a type of supramolecular assembly of two more molecules.

- * The assembly consists of two molecules, electron acceptor and electron donor, that self-attract through electrostatic forces.

- * The conductivity of CT complex is dependent on the ionic binding between the donor & acceptor components.

- * The formation of a conductive CT complex can be employed to design molecular electronic devices.

ex:- gold nanoparticle-polymer complex memory devices

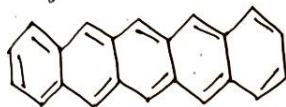
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2. Explain the types of Organic memory devices by taking p-type and n-type Semiconductor materials.

Ans:- Organic memory devices are the memory devices which are fabricated with organic semiconductor (p-type) or (n-type).

P type semiconductor in Organic memory device

* Pentacene is an example for p-type semiconductor structure of pentacene is.



* For constructing memory device, pentacene is typically sandwiched between two electrodes.

Cathode and anode

* Pentacene acts as a active layer in memory device when a voltage is applied across the two electrodes, electrons flow from the anode into the pentacene layer, where they are accepted by the pentacene molecules.

* The flow of electrons causes a change in the electrical properties of the device, this change can be used to represent a bit of information, such as Zero or One.

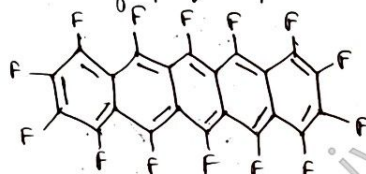
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* The state of the memory device can be read by applying a smaller voltage across the two electrodes and measuring the resulting current.

n-type semiconductor in Organic memory device.

* Perfluoropentacene is an example for n-type Organic memory device.

Structure of perfluoropentacene is



* For constructing memory device, perfluoropentacene is typically sandwiched between two electrodes Cathode and anode

* These perfluoropentacene molecules act as the active layer in the memory device.

* When a voltage is applied across the two electrodes the cathode releases electron from perfluoropentacene layer

* The flow of electrons causes a change in the electrical properties of the device, This change can be used to represent a bit of information, such as Zero or One.

* The state of the memory device can be read by applying a smaller voltage across the two electrodes & measuring the resulting current.

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Q4. What are photoactive materials and Electroactive materials. Explain the working principle of these in Display system.

Ans:- Photoactive Materials

- * Photoactive materials are those that can absorb light and convert it into an electrical signal.
- * These materials are used in photo sensor & photovoltaic cells.
- * When a light falls on a photoactive material, it releases electrons which generate an electric current.
- * Photoactive materials are used in LCD panels for controlling the pixels in the display.

Working principle

- * In LCD display photoactive materials are used to control the pixels by allowing or blocking the light.
- * A backlight is placed behind the LCD panel, which emits light that passes through a polarizing filter.
- * Photoactive materials are placed in front of the polarizing filter.
- * By applying an electric field to the photoactive material, there will be a change in the orientation of the molecules within the material.
- * This changes the amount of light that is allowed to pass through the material, thus controlling the brightness of the pixel.



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Electroactive material

- * Electroactive materials are those that can change their physical properties such as shape, size or refractive index when exposed to an electric field.
- * These materials are used in electroluminescent displays, which are commonly found in watches, calculators and other portable electronic devices.
- * When an electric field is applied to an electroactive material, it emits light which is used to create the image on the display.

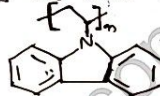
Working principle

- * In display system these materials are usually sandwiched between two conductive layers, which act as electrodes.
- * When a voltage is applied between the electrodes, the electroactive material emits light which is used to create the image on the display.

Q. What are nanomaterials? Explain the ~~the~~ any four properties of poly[9-vinylcarbazole] (PVK) suitable for optoelectronic devices.

Ans:- Nanomaterials are any substances in which at least one dimension is less than 100nm.

Poly (N-vinyl carbazole) or PVK is one of the highly processable polymers as hole conducting material. Structure is



Properties

- * Luminescence:- when excited with light or an electric field, PVK emits light in the blue-violet range of the visible spectrum.
- * Processability:- PVK is a highly processable polymer that can be easily spin coated or printed to form thin films for the application in optoelectronic devices.
- * Solubility:- PVK is soluble in a wide range of solvents, which makes it easy to prepare solutions of the polymer for processing.
- * Thermal stability:- PVK has a high glass transition temperature, and hence it can withstand at high temperatures without degrading.

Applications

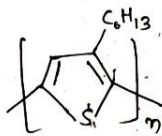
- * Used in the fabrication of light emitting diodes and laser printers.
- * Used in fabrication of organic solar cells when combined with TiO_2 on glass substrate.
- * Used in the fabrication of solar cells when combined with perovskite materials.

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Q. What are optoelectronic devices? Explain the properties of P3HT for optoelectronic devices.

Ans:- Optoelectronic devices are electronic devices that are capable of emitting, detecting or manipulating light.

Structure of P3HT - Polythiophenes $\rightarrow$ light absorbing material



Properties

- * P3HT has great capability as light-absorbing materials in organic electronic devices.
- * P3HT has a ~~very~~ crystalline structure & good charge-transport properties required for optoelectronics.
- * P3HT has a direct-allowed optical transition with a fundamental energy gap of 2.14 eV.
- * Fundamental band gap of P3HT is 490nm visible region, corresponding to $\pi \rightarrow \pi^*$ transition, giving electron-hole pair.

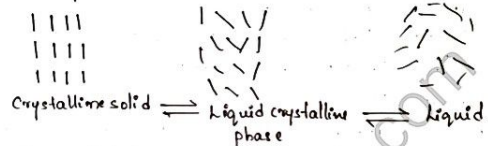
Application

- * Used in the fabrication of Organic field-effect transistors.
- * Used as a hole-transporting layer in Organic light-emitting diodes to improve their efficiency & stability.
- * Used in the construction of Organic solar cells.
- * Used in manufacture of smart windows.

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Q7. What are liquid crystals? Explain the classification of liquid crystal.

Ans: Liquid crystal (LC) is a state of matter whose properties are in between conventional liquids and solid crystals.

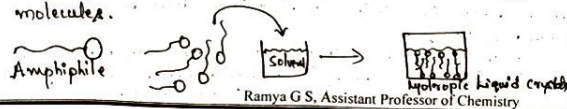


Liquid crystals are classified into two main categories

1. Lyotropic liquid crystals
2. Thermotropic liquid crystals
 - (i) Nematic LC
 - (ii) Smectic LC
 - (iii) Cholesteric LC

1. Lyotropic liquid crystals

- * The orientational behaviour of lyotropic crystals is a function of concentration and solvent.
- * These molecules are amphiphilic - they have both hydrophilic and hydrophobic ends in their molecules.



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- * The hydrophobic end is water repellent whereas the hydrophilic end is attracted towards water and hydrophobic ends are attracted towards non-polar solvents.

- * At low concentrations, these molecules are randomly oriented but as the concentration increases, the molecules start arranging themselves.
- ex: Soaps & detergents form lyotropic crystals when they combine with water.

2. Nematic LC

- * The molecules are oriented along a common axis but are free to rotate around it.
- * Used in LCDs

3. Smectic LC

- * The molecules form layers with their long axes aligned parallel to one another
- * Used in nonlinear optics

4. Cholesteric LC

- * The molecules are arranged in a helical structure
- * Used in liquid filters and reflectors

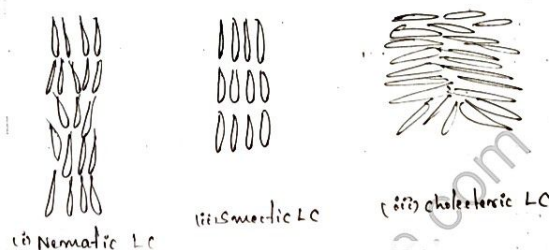
Nematic, smectic and cholesteric LC are called thermotropic liquid crystals, because the molecular orientation in this crystal is depends on the temperature.

By increasing the temperature, the increase in energy and thereby movement of constituent molecules, will induce phase changes.

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Q7.

Crystal $\xrightarrow{\text{Heat up}}$ Thermotropic liquid crystal $\xleftarrow{\text{cool down}}$ Liquid. (i)



Q. Mention any four properties and applications of OLED.

Organic light emitting diodes (OLEDs) are organic thin films which are sandwiched between two electrodes and capable of emitting light when an electric current is applied to them.

Properties:-

- * OLEDs are very thin and flexible, which makes them suitable for use in curved or flexible displays.
- * OLEDs have high contrast ratio, which means that they can produce deep blacks and bright whites, resulting in images with vivid and rich colours.
- * OLEDs have a fast response time, which means that they can switch on and off quickly, resulting in smooth and seamless motion in video content.
- * OLEDs have a wide viewing angle, which means that the image quality is maintained even when viewed from different angles.

Applications of OLED

- * OLED displays are used in televisions, monitors, smartphones and other electronic devices.
- * OLEDs can also be used as a source of lighting in various applications.
- * OLEDs are used in automotive applications, such as dash-board displays, interior lighting and taillights.
- * They are used in medical applications, such as surgical lighting & medical imaging.

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Q.4. Mention any 4 properties and applications of QLED.

Ans:- Quantum dot light-emitting diodes (QLEDs) are a type of electroluminescent devices that utilize quantum dots such as cadmium selenide or zinc sulphide nano crystals.

Properties

- * QLEDs are capable of producing highly accurate and vibrant colours.
- * QLEDs are more energy-efficient than traditional LCD displays.
- * QLED displays have high contrast ratios compared to LED's.
- * QLED displays have fast response times, compared to LED's.
- * QLEDs have a longer lifespan than traditional LCD display.

Applications

- * Used in televisions and displays.
- * Used as a source of lighting in various applications.
- * Used in medical imaging applications, such as in MRI machines, to produce high resolution & accurate images.
- * Used in advertising displays, such as digital billboards to produce high-quality & eye-catching visuals.

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